



Guidelines for Writing Graduation Research Project

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A STEP-BY-STEP GUIDE
for Writing Graduation Research Project

Table of Contents

Table of Contents, List of Tables/Figures, List of Abbreviations	1
Research article	2
Literature Review Article	3
Systematic Review Article	4
References and General Considerations	5
General instructions	6

Title page, Declaration, Dedication, and Acknowledgment

TITLE PAGE

The title page is the first page of your document and includes logos of Higher Education and the College of Pharmacy and essential information such as the title of your work, your name, and the institution you're affiliated with. It sets the stage for your document and provides a formal introduction.

DECLARATION

The declaration is a statement where you affirm that the work is your own and that you have not plagiarized any part of it. It often includes a signature and date, confirming the authenticity and originality of your work.

DEDICATION

The dedication page is where you can honor someone special by dedicating your work to them. This could be a family member, friend, mentor, or anyone who has inspired or supported you. It's usually a short, personal note.

ACKNOWLEDGMENT

The acknowledgment section is where you thank those who helped you while creating your work. This can include advisors, colleagues, friends, and family. It's a way to thank those who contributed to your success publicly.

Note

Each section, including the Title Page, Declaration, Dedication, and Acknowledgment, should be presented on its own separate page.

Abstract section

HERE'S HOW IT WRITES

A one-page abstract of no more than 250 words is required. Five keywords should be provided by the researcher at the bottom of the page. It can be written as a single paragraph for a review article and should not include any references or abbreviations.



In a research article, the abstract should follow this format, which often includes four subheadings:

Background and objectives: This section provides an overview of the study, its reasoning, and its purpose, as well as a brief problem statement.

Methods: Experimental, observational, and analytical methodologies, the study's design and context, fundamental processes, subject and control animal selection, and experimentation are all part of this.

Results: It contains the primary results, including details about the data and their statistical significance.

Conclusion: Key findings (highlighting novel and significant features of the study) and implications are part of it.

Table of contents, List of Tables/Figures, List of Abbreviations

TABLE OF CONTENTS

Chapter headings, section names, first-level subheadings and subsections, and appendices should all be present, with page numbers aligned to the right of the page.



LIST OF TABLES/FIGURES

Along with the names of the tables and figures, these should have the page numbers that correspond to them placed on the right side of the page.



LIST OF ABBREVIATIONS

Any acronyms or abbreviations used often throughout the text that the student fails to define when used are included in the nomenclature. Sort them alphabetically and use only accepted abbreviations. When a term is first used in the text, it is proper to provide both the full form and the abbreviation in parentheses.



A. Research article

1. INTRODUCTION

The introduction should provide an overview of the research problem, a justification statement, aims, and objectives of the study.

AN OVERVIEW OF THE RESEARCH PROBLEM

It includes relevant background information that is current enough for the reader to grasp the context, citing only the most relevant sources without providing a thorough examination of the topic. Therefore, the researcher should omit specifics regarding the data or findings from the referenced paper.

A JUSTIFICATION STATEMENT

The applicant has to show, with logic and proof, that there is a need for this particular study, and that it has the potential to either fill in knowledge gaps or advance substantive or theoretical understanding. Why is this research necessary, and what will its significance be? These are concerns that the rationale should attempt to address.

AIMS AND OBJECTIVES OF THE STUDY

Aim is a broad phrase that shows the research's general objective or intentions, or the study's final overall purpose. This is the ultimate goal of the investigation or the information the researcher is seeking. Arranged list, ideally with a distinct "subheading," is a common way to express objectives. Two or three stated goals are the maximum appropriateness for a research project; in most cases, a single, well-defined aim is acceptable. The introduction takes up about three pages of the total document.



2. CHAPTERS OF THE PROJECT

GENERAL CONCIDERATIONS

Capitalizing the chapter title is necessary. It would be helpful to assign a number to each chapter, e.g. Chapter 1. LITERATURE REVIEW

- The first letter of the sub-section's title should be capitalized.
- There ought to be a numerical index for every subsection of a chapter, e.g. the first section of chapter 1 is written as 1.1.
- Further subsection/subheading of the above is written as 1.1.1.
- The next section will be 1.2. and for further subsections as 1.2.1.
- Having more than three levels will make the text messy, so the researcher should avoid them.

Chapter 1: LITERATURE REVIEW

It is important that the literature review be current and include a summary of the most important studies on the topic. Current research with a regional or local focus should take center stage. The review should primarily focus on current literature and papers published within the past five years.

A literature review is more than just a citation of sources or an inventory of all the references you can find. Critical synthesis, on the other hand, is the only way to go after conducting extensive research and contemplation of the relevant material. Rather than being a simple overview, it needs to be an analysis and critique of research that tackles a specific issue. To show that he has a better grasp of the subject, the researcher should add his thoughts and commentary to the studies wherever possible.

An effective review follows a historical format that highlights the development of the field over time and includes the location of the studies, both of which provide depth and relevance to the review.

It is not necessary to go into great detail about each and every study; rather, just those works that are regarded as "landmark" in the worker's field of study should be covered. Actually, it is more effective to create a more insightful and instructive line first, and then cite all the sources that supported it.

To make this chapter easier to read and understand, it should be structured with sections, headings, subsections, and subheadings that cover relevant topics such as: terminology and concept definitions; historical context; anatomy and physiology; epidemiological and/or clinical considerations; diagnostic and laboratory techniques; and so on.

A lengthy literature review will not strengthen a project that is not strong overall, so keep this chapter to around 25% of the total length of the thesis project. Any given topic could have different references cited in this chapter. Neither under-nor over-referencing is acceptable.



Chapter 2: METHODS

MATERIALS AND METHODS STUDY

When studying humans or other animals, or doing clinical or laboratory procedures, this is known as a materials and methods study.

PATIENTS AND METHODS

when the research involved human subjects exclusively.

SUBJECTS AND METHODS

when all participants were in good health.

The researcher's materials and techniques should be described in sufficient detail so that other researchers can replicate the approach (s).

This chapter should include:

Full description of:

1. Design of the study.
 2. Setting (place of the study area including description of a hospital).
 3. Time of the study.
 4. Patients, control groups, or laboratory animals will be chosen based on inclusion and exclusion criteria, as will the target population, sampling strategy, and estimated sample size.
- 



RESEARCH ETHICS

Both the college's research ethics committee's written clearance and the participants' informed consent are typically necessary for any study to proceed.

This section details the techniques, instruments, equipment, kits (with the names and addresses of the makers), questionnaire design, and data collection methods. Brief descriptions of well-established methods or procedures are appropriate, while more thorough descriptions, accompanied by references, are required for less well-known yet published approaches. Additionally, thoroughly describing new or significantly altered approaches is essential.

Detailed naming of all chemicals and pharmaceuticals, including their generic names, dosages, and administration routes, as well as the names and addresses of their respective makers.

Statistical terminology and abbreviations should be defined, and there should be a brief summary of the procedures employed and the statistical significance of the results (if applicable). Please specify the computer program that was utilized.

Note

It is more appropriate to place detailed statistical analyses, surveys, scores, maps, etc., in the appendix. Subheadings and subsections are additional acceptable ways to organize this chapter.



Chapter 3: RESULTS

Present the results in a logical order using text, tables, and figures; highlight the most key findings first; provide no commentary. Use subsections and subheadings as needed to divide them into logical sections.

The absolute values and derivatives (such as percentages or rates) of numerical results provide appropriate ways to present them. Specify the statistical procedures used to examine the findings. Except for the appendices, large tables are not appropriate. Splitting the table into two or more smaller ones should be the first priority if it becomes too large to fit on an A4 sheet.

An excellent way to present findings effectively in research writing is to use a storytelling style. This is especially true in the Results section, where the information flows from broad to specific. Before going into the specifics of the study's results, this format gives readers a good grasp of the bigger picture. Here's how this approach works:

A BRIEF SUMMARY OF THE STUDY'S FINDINGS SHOULD BE YOUR FIRST STEP:

Without immediately drowning the reader with specifics, this provides a bird's-eye view of the overall results.

TRANSITION TO KEY FINDINGS:

Progressing from a high-level overview to more specific findings help set the stage.

PROVIDE SPECIFIC DATA:

After the main findings have been presented, go into the details of the data and statistics. Statistical measures, such as p-values or numerical findings, can be included in this.



HIGHLIGHT SUBGROUP OR SPECIFIC ANALYSES:

Additional results, such as subgroup analyses or comparisons between groups, might be considered as you delve deeper into the data. By providing readers with these detailed insights, you let them understand how various elements in your study could impact the final results.

The purpose of presenting the data in a table, diagram, or text relies on the message you wish to convey. You cannot discount the merits of any one data visualization method.

DUPLICATION OF INFORMATION IN TABLES AND FIGURES IS NOT ACCEPTABLE

Tables and figures should be made as self-explanatory as possible, since many will be used directly in the presentation. Statistical measures of variations and tests of significance with significance values should be included in the table.

VERB TENSE IN RESULTS SECTIONS

Past tense (either active or passive voice) is used when the sentence focuses on

the completed study: what was done and found.

The present tense is used:

- To describe an “always true” situation; and
- When the sentence focuses on the document, which will always be there.

e.g. The effect of lncRNA expression on the metastasis of breast cancer patients is summarized in Table 1.



Figure Notes

The word "figure" is used for graphs, diagrams, charts, illustrations, sketches, photographs, etc. The abbreviation "Fig." may be used. Whatever the choice of the researcher, it should be consistent throughout the study. The letter – F – should always be written in Capital.

Table Notes

The word "table" should be written without abbreviation, and the letter – T – should always be capitalized.

- Data in tables or figures should not be fully repeated in the text; only important observations should be emphasized or summarized.
- In tables, each column and each row should have a short or abbreviated heading with a statement of the units of measurement, if applicable. There should be no internal horizontal or vertical lines. Explanatory matters for abbreviated headings should be written in footnotes.

Figures & Tables Note

Tables and figures should be numbered consecutively in the order of their first citation in the text, and they should have a brief informative title or legend. Another type of numerical index for both tables and figures can be used, referring to the chapter number before the table or figure number.

- The title of the table is written above the table while the legend of the figure is written below.
- Both tables and figures should appear in the text after they are first mentioned. Appending them at the end of the chapter in which they are cited makes the research unpleasant to read. The raw data may be given as an appendix.



Chapter 4: DISCUSSION

Discussion is a critique of what was done and what was found. It may start with a concise restatement of the study's purpose and the answers that the study has given.

The flow of ideas in this chapter should follow the sequence of the objectives and results.

Chapter 5: CONCLUSIONS AND RECOMMENDATIONS

This chapter should include a summary of new observations, new interpretations, and new insights that have resulted from this work. Often, the researcher uses this chapter to merely restate the research findings.

He should write real conclusions and implications, making the reader understand what the findings mean and what are the key ideas that can be drawn from the study.

Similarly, recommendations should derive from the candidate's experience in conducting the research and from the study findings stating gaps not covered by the work or unanswered questions requiring investigation, and those discovered during the conduct of the study.

Both conclusions and recommendations should be written starting with the most important points in descending order of importance, which may be enumerated for clarity.

B. Literature Review article

Writing an Introduction

Writing an introduction to a review article requires a structured approach to establish the context of your review, highlight the significance of the topic, and clarify your review's purpose. Below are the key components, along with an explanation of how to include each:

1. PROVIDE BACKGROUND ON THE TOPIC

This is always the first step because it sets the foundation for the reader. Before diving into the specifics, the reader needs to understand the basic concepts and context. Offering a clear background introduces the topic and prepares the reader for what follows.

Start by introducing the topic broadly. Explain what the subject matter is and provide necessary background information, so readers unfamiliar with the topic can follow. Include relevant definitions, historical developments, and any key concepts that are foundational to understanding the rest of the article.

2. STATE WHY THE TOPIC IS IMPORTANT AND TIMELY

Explain why this topic matters in the current academic or societal landscape. Is it a pressing issue? Are there recent developments that make it particularly relevant now? This helps readers understand the urgency and potential impact of the subject review in present-day research.

3. MENTION THE GAP OR RESEARCH QUESTION

Identify the knowledge gap or key question that your review aims to address. This highlights why your review is necessary. It could be an unresolved debate, lack of consensus, or an area underexplored by previous research. This signals to the reader that there is still work to be done in this area and that your review will contribute to addressing these gaps.



4. DEFINE THE SCOPE AND OBJECTIVE OF THE REVIEW

Be clear about what you will and won't cover, and what the main objectives of your review are. This ensures the reader knows the focus of your article and what to expect.

Clearly state the objective of your review. Are you summarizing recent research? Are you critically evaluating certain studies? Set clear boundaries regarding the aspects of the topic you will cover.

5. MENTION THE STUDY'S IMPORTANCE

Finally, conclude the introduction by emphasizing why your review is valuable to the field. How does your synthesis of current research contribute to better understanding or progress in the area?

This helps the reader understand the contribution of your review and its broader implications, closing the introduction with a sense of purpose.



2. The Body of a Review Article

The body of a review article is the heart of your paper, where you delve into the details of the topic. To write it effectively, it's important to follow a clear structure that guides the reader through your analysis.

A. ORGANIZE THE REVIEW THEMATICALLY, CHRONOLOGICALLY, OR METHODOLOGICALLY

Thematic Organization: Divide the review into subtopics based on different themes or aspects of the topic. This is ideal if there are multiple angles to explore (e.g., clinical applications, biological mechanisms, ethical issues).

Chronological Organization: This approach works well when the development of the research or topic has evolved over time. Highlight how understanding or technology has advanced and what significant changes have occurred.

Methodological Organization: Organize based on different methods or techniques used in the research. This structure is useful when comparing how various approaches have contributed to knowledge in the field.

SUMMARIZE KEY FINDINGS FROM THE LITERATURE

For each subtopic, provide a concise summary of the most relevant studies or findings in that area. Highlight the major contributions to the field and what each study brings to the table.

Ensure you focus on both formative studies and recent research to give a balanced view of the current understanding.

COMPARE AND CONTRAST DIFFERENT STUDIES

After summarizing, it's important to go deeper by comparing and contrasting different studies. How do their methodologies, results, or interpretations differ? Are there conflicting findings, or do they support each other? Discuss any patterns or inconsistencies that emerge from the literature.



IDENTIFY GAPS IN KNOWLEDGE AND AREAS FOR FUTURE RESEARCH

As you review the literature, you will notice areas where knowledge is lacking or where further research is needed. These gaps can include unresolved questions, insufficient data, or emerging trends that require more investigation.

Pointing out these gaps is crucial because it highlights where the field can advance and provides a foundation for future studies.

By following this structure, the review body will present a well-organized, thorough discussion of the literature, making it easy for readers to follow the progression of research and understand the current state of the field.

C. Systematic Review And Metaanalysis

In the second style of writing a review article (systematic review), which includes both a Methodology and a Discussion section in addition to the subtopics, the Methodology becomes essential in providing transparency on how the literature was collected and evaluated. This format is common in systematic reviews or meta-analyses and helps ensure that the review is comprehensive, unbiased, and replicable.

METHODOLOGY SECTION

Clearly define the time frame during which the studies you reviewed were published. Indicate the range of years you considered when collecting studies. You may want to focus on more recent research (e.g., the last 5-10 years) or include a longer timeframe if the topic has a longer history of research.

This helps readers understand the temporal scope of the literature you are reviewing.

Detail the specific search terms or keywords used in the review. These keywords should reflect the major themes of your review, helping ensure you gather relevant papers. List both general terms and more specific terms related to your topic.

DISCUSSION SECTION

The discussion section is where you synthesize and interpret the findings from the literature, offering insights into the implications of the research and suggesting future directions. This section should go beyond summarizing studies; it should analyze and critically evaluate the literature.

Here, you analyze the key results or trends found in the studies. What do these findings mean for the field? Discuss whether the results are consistent across studies or if there are significant differences.

This involves critically evaluating the methodologies and outcomes of different studies. Highlight differences in study designs, sample sizes, or techniques, and explain how these might account for varying results.

Discuss limitations in the existing body of research, such as small sample sizes, short follow-up periods, or lack of diversity in study populations.

Highlight gaps where research is lacking or emerging trends that need further exploration. This helps to justify the need for ongoing research.

Based on the gaps and limitations identified, suggest specific areas where future research could focus. This may involve calling for larger, more diverse trials, or suggesting new approaches to studying unresolved questions.

CONCLUSION SECTION

Writing the conclusion of a review article is essential to summarize the key takeaways, emphasize the importance of ongoing research, and suggest future directions for the field. A strong conclusion should leave readers with a clear understanding of the review's significance and how the topic might evolve.

Briefly summarize the major points or themes discussed in the body of the review. This is a condensed version of your key findings, allowing the reader to quickly recall the most critical insights from your analysis of the literature.

After summarizing the findings, emphasize why further research is crucial. This shows the reader that despite advancements, the topic still requires attention for better outcomes or understanding. Mention any ongoing developments or new approaches that hold promise for the future.

Based on the gaps and limitations discussed in the body, offer specific suggestions for future research. This could include exploring new methodologies, addressing unresolved questions, or investigating emerging trends. These recommendations should be practical and forward-looking, helping guide the next steps in the field.

REFERENCE SECTION

Students can demonstrate their reading comprehension by citing their sources. It is reasonable to accuse the candidate of plagiarism if they do not provide full sources. One of the most serious forms of academic misconduct is plagiarism. The researcher needs to be familiar with the specific rules and disciplinary actions put in place by the HMU to ensure that their work does not plagiarize.

REFERENCE NOTES:

- Use the Harvard style for referencing this piece of work.
- Citations should be based on original research publications published in journals with an index in PubMed or a similar database. Never depend on unpublished notes, private correspondence, or even theses and dissertations as sources. The only time you should acknowledge them is if the information they give is both crucial and hard to get elsewhere.
- Put "In press" after the title of the article if it refers to a published but unpublished work.
- "Unpublished observations" should be cited in the text as information from manuscripts that were submitted for publication but were not accepted.
- Whenever you use a table or figure from another published work, be sure to take permission and cite it.
- Figures are typically copyrighted, even if you cite the original paper. You will need to obtain permission from the copyright holder. Many journals and publishers use platforms like Rights Link to facilitate obtaining permissions for the reuse of figures. Follow the process outlined by the publisher to request permission. Some journals allow reuse under a Creative Commons (CC) license, which might not require permission for non-commercial use, as long as the figure is properly cited. Check the licensing terms carefully.
- Example of how to cite a reused figure in a caption:
- "Adapted from Smith et al., 2021 with permission from [Publisher Name]."
- Some journals have specific policies regarding figure reuse and permissions. Be sure to check the guidelines for the journal where you plan to submit your work.

- Always check both the copyright status and journal guidelines for proper citation and permission processes.
- With six or fewer authors, it is proper to list each one. There should be no more than six authors named; in such cases, et al. should follow.

GENERAL INSTRUCTIONS

- Research writing should be on one side of white paper A4 (297x 210 mm).
- A margin is the space between a document's contents and the edges of the page. The margins of the pages should be in inches [top:1, left:1.5, bottom:1, right:1] or in centimeters [top:2.54, left:3.8, bottom:2.54, right:2.54].
- Use the automatic page numbering.
- The content of the whole report is written in size (12) and font (Times New Roman), except the title section, which should be in bold and the size is (14).
- Number the illustrations according to their sequence in the text.
- All figures in the research work must be original and not copied from online sources. If you use a previously published figure, it is necessary to obtain permission from the main authors or the journal where the figure was published.
- Use of Mendeley or Endnote as a reference manager software is preferable.
- The majority (more than 75%) of the references should be published during the previous 5-10 years of the date of discussion
- Line spacing should be 1.5 spaces of the Microsoft Word computer program throughout the text.
- Pagination in Arabic numerals starts with the first page of the introduction through the recommendation. Page numbers should be at the bottom center of the page. The minimum font size for Arabic numerals is 10, which must be consistent throughout the project.
- The AI content and plagiarism rate will be checked by the department and your project will be accepted based on the Turnitin result.
- Total number of pages is usually 20-25 pages for the research article project and 25 to 35 pages for the review article project.

Successful writing skill is a learnable skill

2024-2025

